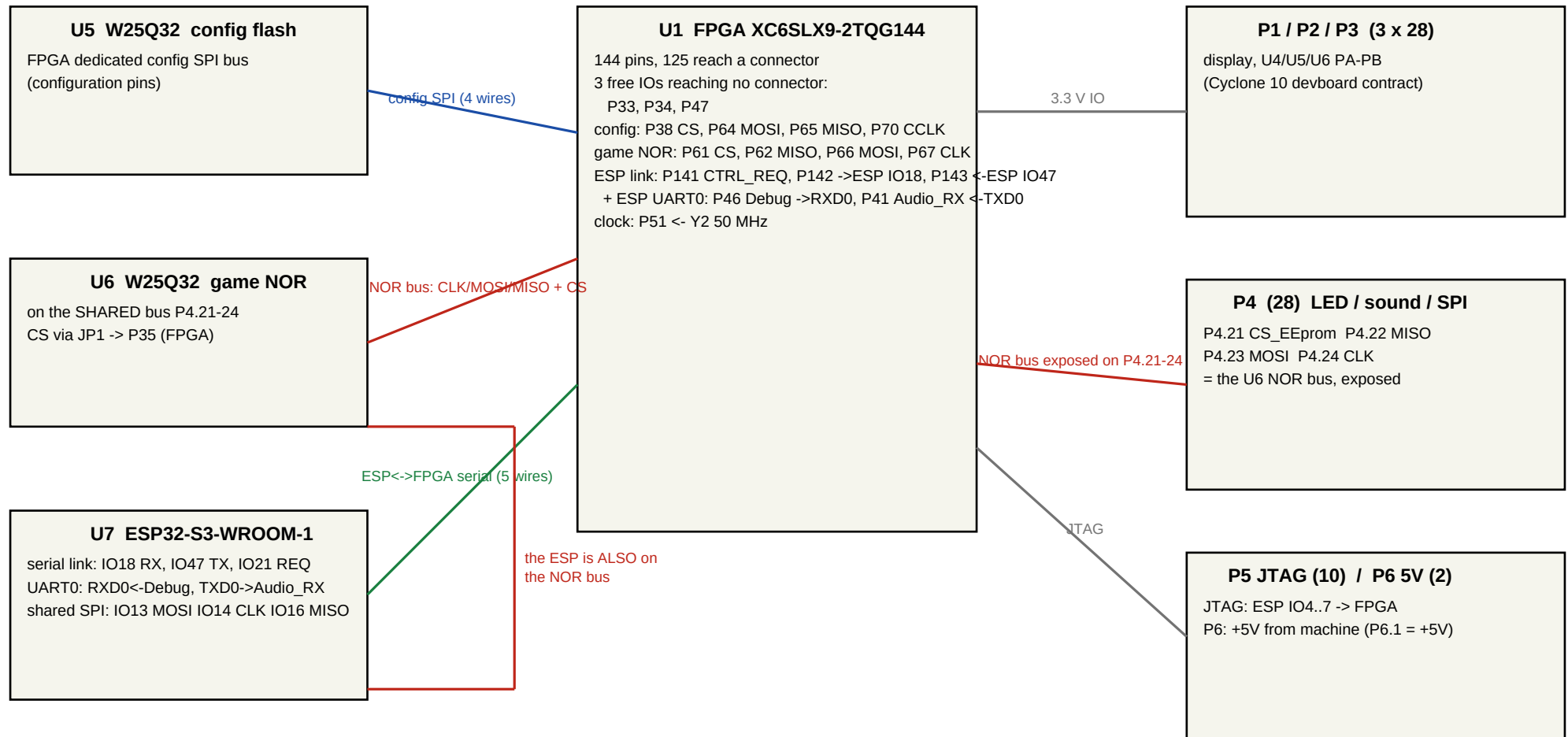


Smart FA (GottFA_SLX9 module) -- bus overview

What each link carries. The following pages wire every connector pin to its FPGA pin.



What this module cannot do, and why (reply to Ralf, 2026-09-07)

- P4.21..24 = the NOR SPI bus, at the GottFA80 positions. On WillFA7 those positions are the DIP strobes: incompatible.
- SPI is not at the same place from one FA board to the next (Ralf): a NOR on the connector can never be universal.
- P2.59, P2.64, P2.65: NOT CONNECTED on this module, and free on every FA board -> the designated place for the ESP<->FPGA serial link.
- For an internal NOR: only 3 FPGA IOs reach no connector (P33, P34, P47) + CS P35. CLK+MOSI+MISO+CS fits, with no spare.
- The ESP would then lose direct NOR access: U6 written through the FPGA (a bridge), as U5 is already programmed through the BSCAN bridge.

P1 -- display

Every pin is wired to its FPGA pin; the other parts on the same net sit along the wire. Power in grey.

P1		U1 FPGA	
1	disp_segments[1]	P74	L74N_DOUT_BUSY_1_74
2	disp_segments[2]	P75	L74P_AWAKE_1_75
3	disp_segments[3]	P78	L47N_1_78
4	disp_segments[4]	P79	L47P_1_79
5	disp_segments[6]	P80	L46N_1_80
6	disp_segments[7]	P81	L46P_1_81
7	disp_segments[5]	P82	L45N_1_82
8	disp_segments[8]	P83	L45P_1_83
9	disp_segments[9]	P84	L43N_GCLK4_1_84
10	disp_segments[10]	P85	L43P_GCLK5_1_85
11	disp_segments[11]	P87	L42N_GCLK6_TRDY1_1_87
12	disp_segments[12]	P88	L42P_GCLK7_1_88
13	disp_segments[13]	P92	L41N_GCLK8_1_92
14	disp_segments[14]	P93	L41P_GCLK9_IRDY1_1_93
15	disp_segments[15]	P94	L40N_GCLK10_1_94
16	disp_segments[16]	P95	L40P_GCLK11_1_95
17	disp_segments[17]	P97	L34N_1_97
18	disp_segments[18]	P98	L34P_1_98
19	disp_segments[19]	P99	L33N_1_99
20	disp_segments[20]	P100	L33P_1_100
21	disp_segments[21]	P101	L32N_1_101
22	disp_segments[22]	P102	L32P_1_102
23	disp_segments[23]	P104	L1N_VREF_1_104
24	disp_segments[24]	P105	L1P_1_105
25	GND		
26	GND		
27	VCC3.3		
28	VCC3.3		

P2 -- U5/U6 port A-B (P2.59/64/65 do not exist on this 28-pin module: see page 1)

Every pin is wired to its FPGA pin; the other parts on the same net sit along the wire. Power in grey.

P2		U1 FPGA
1	GND	
2	GND	
3	unconnected-(P2-Pin_3-Pad3)	
4	U5_PA[0]	P111 L66N_SCP0_0_111
5	U5_PA[1]	P112 L66P_SCP1_0_112
6	U5_PA[2]	P114 L65N_SCP2_0_114
7	U5_PA[3]	P115 L65P_SCP3_0_115
8	SPARE_IO_1	P116 L64N_SCP4_0_116
9	SPARE_IO_2	P117 L64P_SCP5_0_117
10	U6_PB[2]	P118 L63N_SCP6_0_118
11	U6_PB[1]	P119 L63P_SCP7_0_119
12	DIP_Return[3]	P120 L62N_VREF_0_120
13	SPARE_IO_3	P121 L62P_0_121
14	myTest	P123 L37N_GCLK12_0_123
15	U5_PA_7	P124 L37P_GCLK13_0_124
16	U6_PB[3]	P126 L36N_GCLK14_0_126
17	U6_PB[4]	P127 L36P_GCLK15_0_127
18	U6_PB[5]	P131 L35N_GCLK16_0_131
19	U6_PB[6]	P132 L35P_GCLK17_0_132
20	U6_PB[7]	P133 L34N_GCLK18_0_133
21	U6_PA[3]	P134 L34P_GCLK19_0_134
22	U6_PA[1]	P137 L4N_0_137
23	U6_PA[2]	P138 L4P_0_138
24	U6_PA[0]	P139 L3N_0_139
25	U6_PA[6]	P140 L3P_0_140
26	GND	
27	GND	
28	VCC3.3	

P3 -- U4/U6 port A-B

Every pin is wired to its FPGA pin; the other parts on the same net sit along the wire. Power in grey.



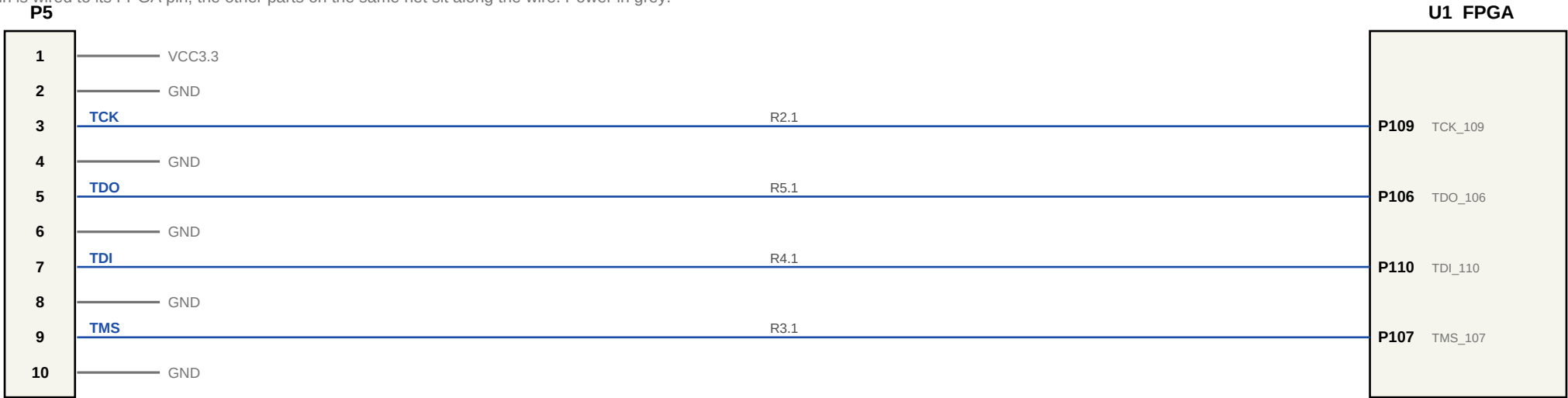
P4 -- LED / sound / SPI

Every pin is wired to its FPGA pin; the other parts on the same net sit along the wire. Power in grey.



P5 -- JTAG

Every pin is wired to its FPGA pin; the other parts on the same net sit along the wire. Power in grey.



Internal links (no connector) -- config U5, NOR U6, ESP link, jumper JP1

The nets that never leave the module. This is where compatibility with the other FA boards is decided.

Config flash U5 <-> FPGA (configuration pins, not exposed)

U1.P38 L65N_CS0_B_2_38	CFG_CS	U5.1 ~{CS}_1
U1.P64 L3N_MOSI_CSI_B_MISO0_2	CFG_DI_MOSI	U5.5 DI/IO_{0}_
U1.P65 L3P_D0_DIN_MISO_MISO1_	CFG_DO_MISO	U5.2 DO/IO_{1}_
U1.P70 L1P_CCLK_2_70	CFG_CCLK	U5.6 CLK_6

ESP32 <-> FPGA serial link (three internal wires -> candidates for P2.59/64/65 on the next board)

U1.P141 L2N_0_141	FA_CTRL_REQ	U7.23 IO21_23
U1.P142 L2P_0_142	ESP32_RX	U7.11 IO18_11
U1.P143 L1N_VREF_0_143	ESP32_TX	U7.24 IO47_24

U6 NOR bus -- SHARED between FPGA, ESP and connector P4 (the WillFA7 conflict)

CLK	P4.24 U1.67 L2P_CMPCLK_2_67 U6.6 CLK_6 U7.22 IO14_22
MOSI	P4.23 U1.66 L2N_CMPMOSI_2_66 U6.5 DI/IO_{0}_ U7.21 IO13_21
MISO	P4.22 U1.62 L12P_D1_MISO2_2_62 U6.2 DO/IO_{1}_ U7.9 IO16_9
NOR_CS_FPGA	JP1.1 U1.35 L1P_3_35
NOR_CS	JP1.2 R26.2 U6.1 ~{CS}_1 U7.8 IO15_8
CS_EEprom	P4.21 U1.61 L12N_D2_MISO3_2_61

Free FPGA IOs reaching NO connector (the only place for an internal NOR)

U1.P34 L1N_VREF_3_34	(not connected)
U1.P33 L2P_3_33	(not connected)
U1.P47 L48N_RDWR_B_VREF_2_47	(not connected)